| INCH-POUND |

MIL-C-55302/59G
31 December 1992
SUPERSEDING
MIL-C-55302/59F
26 August 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, PIN CONTACTS, DECADE INCREMENTS 90, 100 AND 120 CONTACT POSITIONS FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302

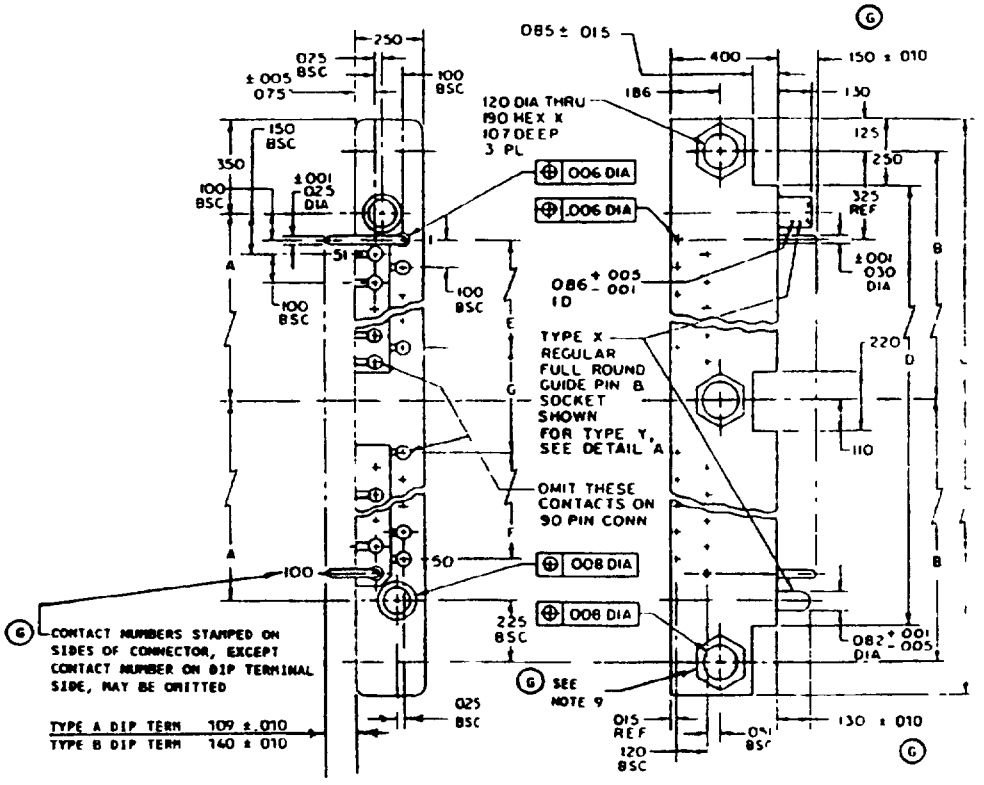


FIGURE 1 Connector, plug 100 (25 4 mm) spacing

(G) Denotes change

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited

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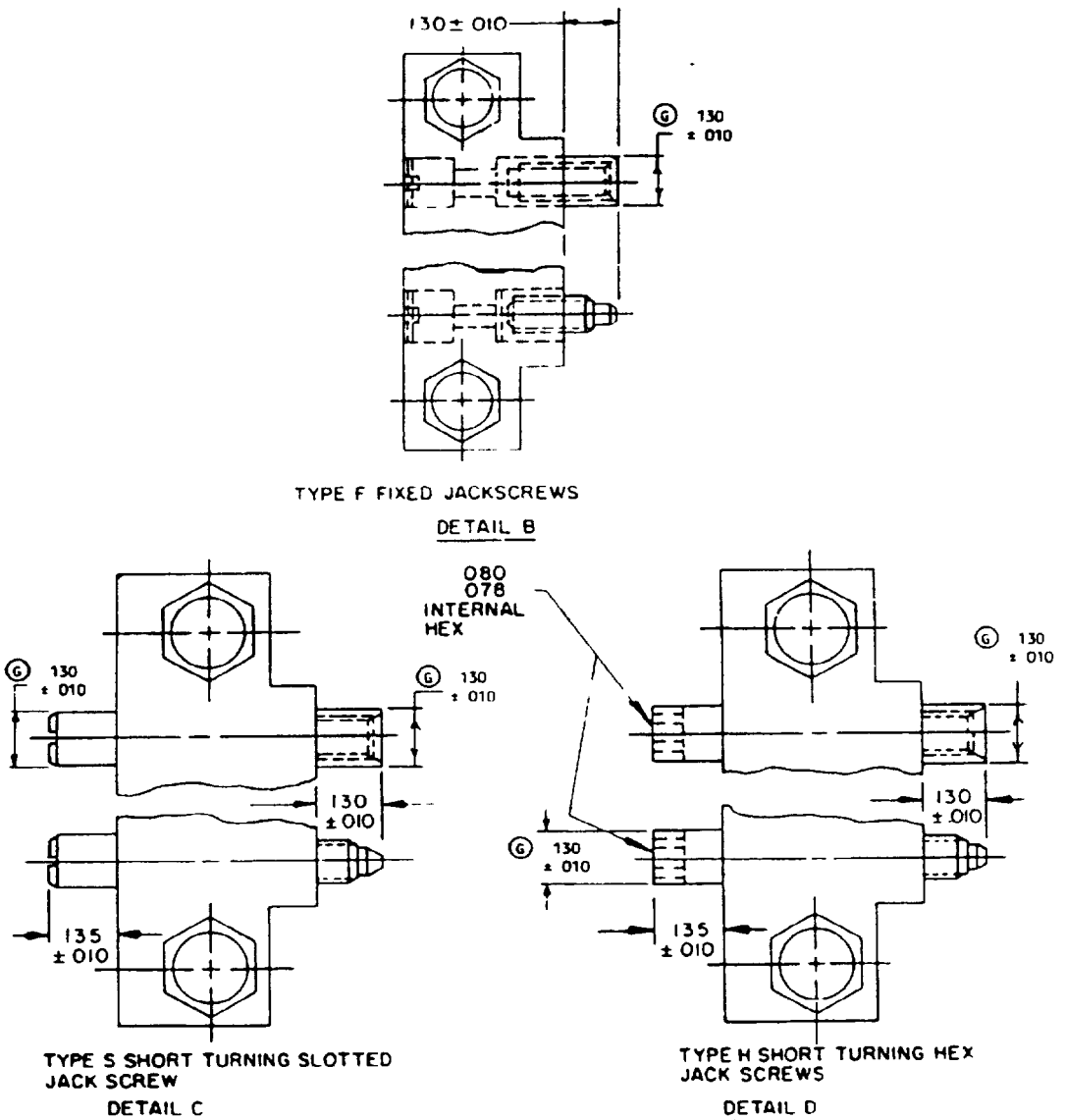
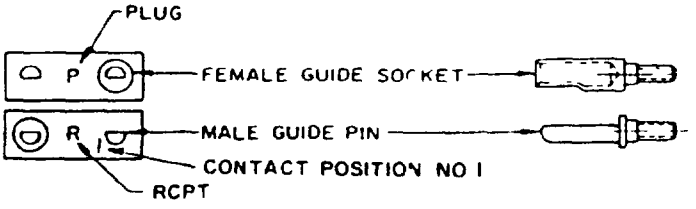


FIGURE 1. Connector, plug 100 (2.54 mm) spacing - Continued.

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CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS (TYPE Y)

DETAIL A

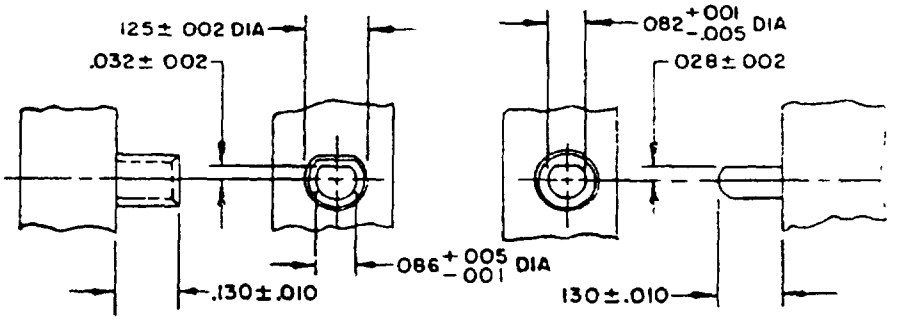


FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
001	0 03	032	0 81	110	2 79	214	5 44
.002	0 05	.051	1.30	.120	3.05	220	5 59
.005	0.13	.075	1.91	125	3 18	.225	5 72
006	0 15	.078	1.98	.130	3 30	235	5 97
.008	0 20	.082	2 08	135	3 43	250	6 35
.010	0.25	.086	2.18	.140	3.56	350	8 89
.020	0.51	.090	2.29	.150	3 81	395	10.03
.025	0 64	.100	2.54	.172	4 37	.400	10 16
028	0 71	107	2.72	190	4 83	425	10 80
030	0 76	109	2 77	200	5 08		

Number of contacts	Dimensions						
	BSC A	BSC B	± 010 C	± 010 D	Ref E	Ref F	Ref G
90	2 475 (62 87)	2 700 (68 58)	5 650 (143.51)	5 150 (130 81)	2.200 (55 88)	2 100 (53 34)	400 (10 16)
100	2 725 (69.22)	2 950 (74 93)	6 150 (156.21)	5.650 (143.51)	2 400 (60.95)	2 400 (60 95)	400 (10 16)
120	3 225 (81 92)	3 450 (87.63)	7 150 (181 61)	6 650 (168.91)	2 900 (73 66)	2 900 (73.66)	400 (10 16)

NOTES

1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ± 005 (0 13 mm)
4 These connectors mate with connectors specified in MIL-C-55302/60
5 Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number 1 contact shall be marked
6 Termination layout on 020 (0 51 mm) modular grid
7 Metric equivalents are in parentheses
(G) 8 Optional undercut 050 (1 3 mm) maximum, on PCB side for cleaning purposes
(G) 9 Position of the hex orientation is optional Hex shall not break through the side of the insulator
(G) 10 Hole depth to female guide hardware is 282 min (7 16 mm) Full thread depth to female threaded hardware is .240 min (6 1 mm)

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued

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REQUIREMENTS

Dimensions and configuration See figure 1

Material Guide pins, type X guide sockets, jackscrews and jacksockets shall be made of corrosion resisting stainless steel in accordance with ASTM-A581 or ASTM-A582 passivated in accordance with QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2.5% max other, and the balance zinc) and shall have a brinell hardness of 145 to 175 Type Y guide sockets shall be beryllium copper in accordance with QQ-C-550, nickel plated in accordance with QQ-N-290, class 2, 50 to 150 microinches.

Plating. The contact plating shall be in accordance with MIL-C-55302 or gold in accordance with MIL-G-45204, type II, grade C, class 1, over nickel plating in accordance with QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating in accordance with QQ-N-290, class 02, 50 to 150 microinches in the termination area

Contact identification See figure 1

Contact resistance The average contact resistance of all contacts measured shall not exceed 0 010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm

Dielectric withstanding voltage

Sea level· 1,000 volts rms
High altitude. 300 volts rms

Current rating 5 0 amperes maximum per contact

Polarization: Type X, regular, guide pin and socket, full round (see figure 1) Type Y, adjustable "D" shaped guide pin and/or socket (see MIL-C-55302/57, detail B) All connectors are supplied in the -1 polarized position. For other than OEM use, connectors may be re-polarized by using M55302/57-01 spanner wrench and M55302/57-02 guide pin and guide socket set. When guides are removed and reinstalled in another position apply Loctite 83-21 or equivalent to the threads Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three spaces after the part number on the purchase order Example M55302/59-A10Y -32 Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench M55302/57-01 Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three spaces after the part number on the purchase order. Guide sets, as installed, shall withstand 7 inch ounces backoff torque applied through the guide set mounting nut

M55302/57-02 kit: Consists of one type Y "D" shaped guide socket and one type Y "D" shaped guide pin

Example of part number

M55302/59	A	100	X
Basic number of specification sheet	Type of terminal (see figure 1)	Number of contacts (see figure 1)	Type of mounting hardware (see details A,B,C, and D of figure 1)

A = dip terminal .109 long
B = dip terminal 140 long
C = dip terminal .172 long

Type X = full round
Type Y = "D" shaped
Type F = fixed jackscrew
Type S = 135 slotted turning jackscrew
Type H = 135 hex turning jackscrew

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Supersession data See table I

(G) TABLE I Supersession data

New part number	Superseded part number
M55302/59*****	M55302/59-*****

CONCLUDING MATERIAL

Custodians.
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3860-5)

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